

224 Gbaud On-Off Keying Optical Link Using O-Band Vertical P–N Junction Ring Resonator Modulator

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Abstract An O-band vertical p–n junction ring resonator modulator-based IM/DD link is demonstrated using bi-LSTM equalization. We achieve 224 Gbaud OOK and 140 Gbaud PAM4 over 1 km single-mode fiber with BER below the 6.25% overhead HD-FEC threshold. ©2026 The Author(s)

Introduction

The rapid growth of data center traffic is driving the demand for higher-speed and lower-power optical interconnects. Silicon photonics (SiP) has emerged as a promising platform due to its CMOS compatibility and scalability [1]. Among various SiP transmitters, ring resonator modulators (RRMs) are attractive for intensity modulation and direct detection (IM/DD) links due to their compact footprint and low driving voltage [2,3]. However, RRM require thermal tuning, and their effective electro-optic (EO) bandwidth becomes a bottleneck at high symbol rates.

Recent work on a vertical p–n junction (VPN) RRM has improved the modulation efficiency and effective EO bandwidth in the O-band [4,5]. However, as the symbol rate increases, RRM-based IM/DD transmission suffers from strong resonance-induced inter-symbol interference (ISI), nonlinear distortion, and chromatic dispersion (CD) [5,6]. These impairments limit the achievable symbol rate and transmission distance, even when using conventional digital signal processing (DSP) methods such as feedforward equalizers (FFE), decision feedback equalizers (DFE), and Volterra filters [7,8]. Using FFE+DFE equalization, the reported performance of this O-band VPN RRM link was 200 Gbaud on–off keying (OOK) and 140 Gbaud four-level pulse amplitude modulation (PAM4) over 500 m single-mode fiber (SMF) [4,5].

Neural network (NN)-based equalization has recently been investigated for high-speed IM/DD links [9–11]. By learning temporal dependencies of sequences, NN equalizers can effectively compensate for nonlinear distortion and long memory

effects. In this work, we apply a bi-directional long short-term memory (bi-LSTM) equalizer to an O-band VPN RRM-based IM/DD link. The bi-LSTM equalizer extends the achievable transmission capability beyond conventional DSP equalizers. We achieve 224 Gbaud OOK and 140 Gbaud PAM4 transmission over 1 km SMF with bit error rate (BER) below the 6.25% overhead (OH) hard-decision forward error correction (HD-FEC) threshold of 4.5×10^{-3} .

Experimental Setup

The experiment uses an O-band SiP RRM fabricated using the imec iSiPP200 platform [12]. The RRM uses a VPN design and has a ring radius of 5 μm with a 140 nm bus-to-ring coupling gap and a 450 nm waveguide width optimized for low insertion loss. The device exhibits an off-resonance insertion loss of 9.2 dB, including 0.7 dB from the RRM and 8.5 dB from grating couplers (4.25 dB per coupler at 1310 nm). The RRM has a modulation efficiency of $V_{\pi}L = 0.34 \text{ V}\cdot\text{cm}$ and a Q-factor of 4958 at 0 dBm input optical power and 1310 nm.

The experimental setup is shown in Fig. 1(a). At the transmitter side, a pseudorandom binary sequence (PRBS) exceeding 10^6 bits is generated offline in MATLAB using a Mersenne Twister generator with a shuffled seed. The sequence is mapped to OOK or PAM4 symbols, upsampled to four samples per symbol, and pulse-shaped using a root-raised cosine (RRC) filter with a roll-off factor of 0.1. The waveform is then resampled to 256 GSa/s to match the arbitrary waveform generator (AWG, Keysight M8199B), which provides

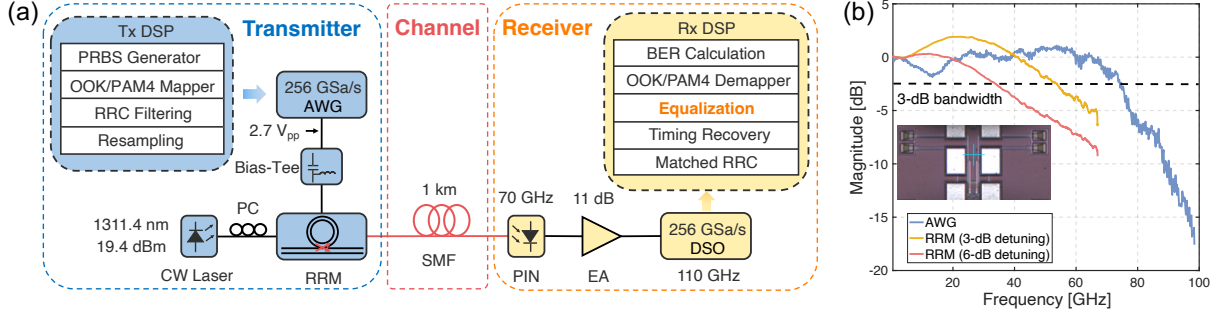


Fig. 1: (a) Experimental setup and (b) measured response of AWG and O-band VPN RRM. Inset: chip photo [13].

a 2.7 V_{pp} output swing. A continuous-wave (CW) tunable laser (Keysight N7776C-013) operating in the O-band provides 19.4 dBm optical power at 1311.4 nm. The RRM is reverse-biased at -2.5 V through a 110 GHz bias-tee (SHF BT110R) and driven by the AWG via ground-signal-signal-ground (GSSG) RF probes, where the SG pair is used for heater tuning. Fig. 1(b) shows the measured frequency response of AWG and the VPN RRM. The RRM exhibits a measured 3-dB EO bandwidth of 55 GHz at 3-dB detuning and 36 GHz at 6-dB detuning. The resonance is tuned to ~4.3-dB detuning to balance the extinction ratio and insertion loss. After modulation, the optical signal is transmitted through a 1 km SMF.

At the receiver side, the signal with up to 4.5 dBm received optical power (RoP) is detected using a 70 GHz PIN photodiode (PD), followed by a packaged electrical amplifier (SHF M827B) with 11 dB gain. The electrical waveform is captured by a real-time digital storage oscilloscope (DSO, Keysight UXR1104A) operating at 256 GSa/s. No waveform triggering or clock synchronization is used between the AWG and DSO to emulate practical data-center conditions. Offline DSP includes matched filtering, timing recovery, bi-LSTM and/or conventional equalization, and BER calculation.

Bi-LSTM Equalizer

The architecture of the bi-LSTM equalizer is shown in Fig. 2(a). The input contains 25 symbols (12 preceding and 12 succeeding). The bi-LSTM layer consists of $n_h = 20$ hidden units (10 forward and 10 backward) with a “tanh” activation function. The hidden state h_i captures inter-symbol dependencies by combining the forward and backward hidden states. The output layer is a fully connected layer with a “sigmoid” activation for OOK and a “linear” activation for PAM4, producing a single output y_i . For PAM4, a k-nearest neighbors (KNN) classifier with $k = 25$ is applied to map the regression output to discrete symbol

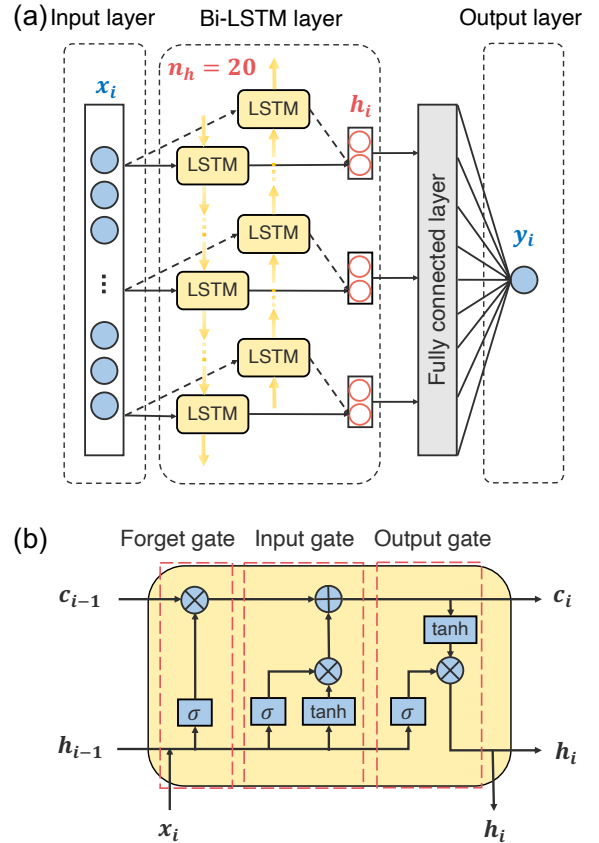


Fig. 2: Architecture of (a) bi-LSTM equalizer and (b) LSTM cell.

levels. The structure of the LSTM cell is shown in Fig. 2(b), where x_i is the current input symbol, c_{i-1} and c_i are the previous and updated cell states, h_{i-1} and h_i are the previous and current hidden states, and σ is the “sigmoid” activation function.

The bi-LSTM equalizer is trained on 2^{17} symbols and tested on 2^{15} symbols for each signal. Input taps, input normalization, activation functions, and batch size are selected using Bayesian optimization. The mean squared error (MSE) is used as the loss function. The Adam optimizer is applied with an adaptive learning rate ranging from 10^{-4} to 10^{-2} . Equalization is performed at

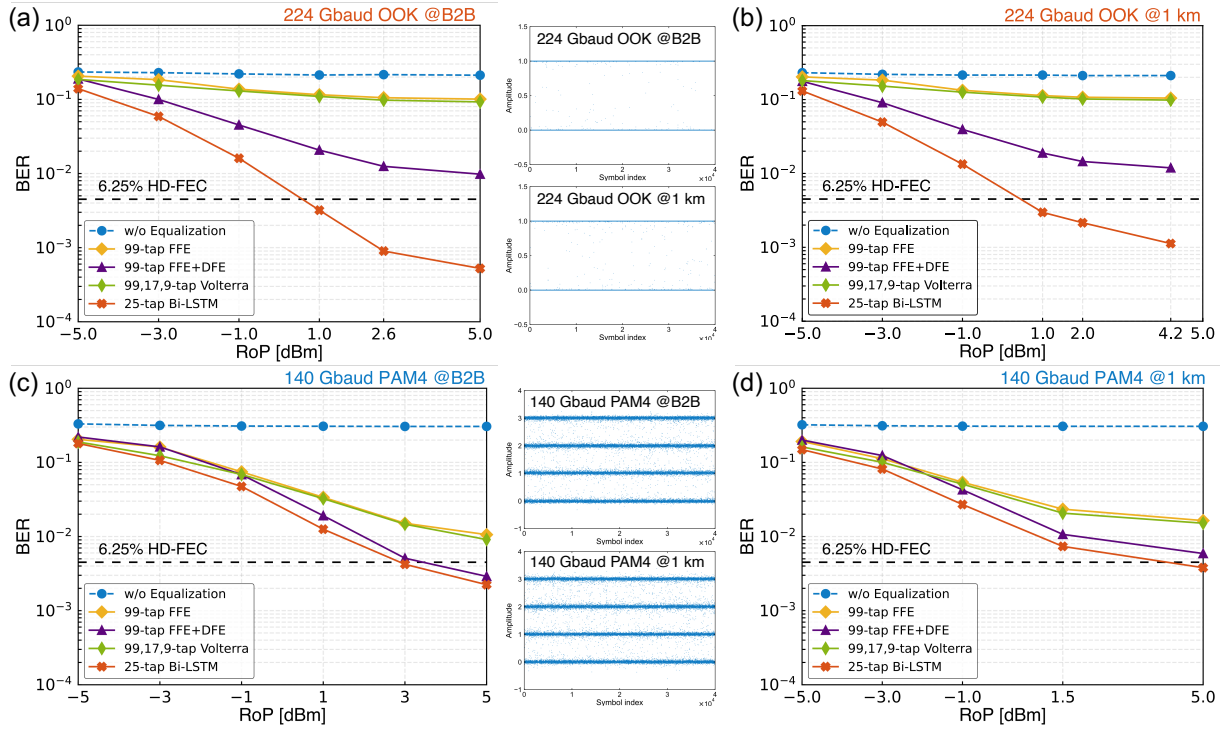


Fig. 3: BER versus RoP for (a) 224 Gbaud OOK in B2B, (b) 224 Gbaud OOK over 1 km SMF, (c) 140 Gbaud PAM4 in B2B, and (d) 140 Gbaud PAM4 over 1 km SMF. Insets: symbol distributions after bi-LSTM equalization.

one sample per symbol. For both OOK and PAM4, the bi-LSTM equalizer is pre-trained on the highest RoP traces for back-to-back (B2B) and 1 km transmission and fine-tuned for other RoP levels using transfer learning.

Results and Discussion

Fig. 3 shows the BER versus RoP performance of different equalizers for 224 Gbaud OOK and 140 Gbaud PAM4 under both B2B and 1 km SMF transmission. Transmission without (wo) equalization is shown as a baseline to illustrate the intrinsic link performance. We also compare the proposed 25-tap bi-LSTM equalizer with conventional 99-tap FFE, 99-tap FFE+DFE, and a (99, 17, 9)-tap third-order Volterra equalizer. The FFE and Volterra provide only limited improvement over transmission without equalization, indicating that feedforward equalization alone is insufficient in this link. By combining the FFE with the DFE (FFE+DFE), the BER is significantly reduced, as the decision feedback can partially mitigate post-cursor ISI. However, FFE+DFE remains above the 6.25% OH HD-FEC threshold except for 140 Gbaud PAM4 B2B. The bi-LSTM achieves the lowest BER and reaches sub-FEC performance at the highest RoPs. The insets in Fig. 3 show the symbol distributions after bi-LSTM equalization at the highest RoPs, illustrating clear symbol separation. After 1 km SMF transmission,

it reaches BERs of 1.11×10^{-3} for 224 Gbaud OOK and 3.80×10^{-3} for 140 Gbaud PAM4.

In [4,5], the same platform demonstrated 200 Gbaud OOK and 140 Gbaud PAM4 transmission over 500 m SMF using FFE+DFE. The improved rate-reach is mainly attributed to increased effective channel memory from resonance-induced ISI together with nonlinear distortion, fiber CD and bandwidth constraints, where conventional equalizers show limited performance. By exploiting bidirectional sequence modeling, the bi-LSTM can better compensate the resulting impairments. The improvement is more significant for OOK transmission because the higher symbol rate imposes a stricter bandwidth limit and stronger ISI. These results demonstrate that the proposed bi-LSTM equalizer effectively extends the transmission capability of the O-band VPN RRM link beyond the limits of conventional DSP equalization.

Conclusions

We demonstrated 224 Gbaud OOK (224 Gb/s) and 140 Gbaud PAM4 (280 Gb/s) transmission over 1 km SMF in an O-band VPN RRM link. The proposed bi-LSTM equalizer enables operation beyond the performance limitations of conventional DSP equalizers. These results highlight the potential of NN-based equalization for high-speed, short-reach optical interconnects.

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